

CSC 1201 Probability and Statistics – Assignment 2

April 7, 2026

Instructions

- Work in groups of no more than five (5) members.
- One member should submit the work on behalf of the group.
- The completed work shall be presented in class. **Any member who misses the presentation shall forfeit all marks awarded for the assignment.**

1. Let X be a continuous random variable uniformly distributed on the interval $[a, b]$, with probability density function

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Prove that

- i. $\mathbb{E}[X] = \frac{a+b}{2}$.
- ii. $\text{Var}(X) = \frac{(b-a)^2}{12}$.

(b) A thin copper wire carries a current X , measured in milliamperes, where

$$X \sim U(4.9, 5.1).$$

Determine

- i. the expected current.
- ii. the standard deviation of X .

2. A discrete random variable X has a probability mass function of the form

$$f(x) = \begin{cases} l(8-x), & x = 0, 1, 2, 3, 4, 5, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the constant l .

(b) Sketch $f(x)$.

(c) Find $P(X > 2)$.

(d) Find the expected value $\mathbb{E}[X]$.

(e) What is $V(X)$?

3. The joint density function of the continuous random variables X and Y is given by

$$f(x, y) = \begin{cases} \frac{6}{5}(x^2 + 2xy), & 0 < x \leq 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine the marginal density function of X .

(b) Compute the marginal density function of Y .

(c) Determine whether X and Y are independent.